11 kV FROM HV SWITCHBOARD
F23-SW-016
2000-F23-SLD-EL-10002

F72-SR-001
415V REAGENTS SWITCHROOM

F72-TR-072
TRANSFORMER ROOM

F72-TR-072
11/0.433kV
1600kVA
Dyn11
z=6%
ANAF
NOTE 3

SWR EARTH BAR
Tx EARTH BAR
2500/ 1A
2.5VA
5P20

F72-SW-072
415V REAGENTS 415V MOTOR CONTROL CENTRE

MCC EARTH BAR
NEUTRAL BAR
NOTE 1
1/700A
2500/ 5A
2.5VA
5P20
TEST
TEST
00F-01
2500/ 5A
2.5VA
C10.5
TEST
PM
20A
63A
2A
32A
240VAC SPRING
CHARGE SUPPLY
00F-08
ACB
2A
AFD
2500A
50G
50
51
50G
51G
24VDC CONTROL POWER SUPPLY
NOTE 7
125A
160A
SD
NOTE 6

E-SHUT
F72SR001-LCS
NOTE 12

CONTINUED ON SHT 2
2000-F72-SLD-EL-10202

Approved	X
Approved as Noted	
Not Approved	
For Information Only	

ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES UNDER THE PURCHASE ORDER / CONTRACT

415V 3NE~ 50Hz 2500A 65kA for 1sec

2F-01	2F-05	1F-05	1F-09	5F-06	5F-01	2F-09	2F-13	1R-13	5F-15	5F-20	3F-01
20A	20A	20A	20A	250A	xxxA xxxA	20A	20A	630A	20A	20A	20A
2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A	2A 32A
70A	70A	70A	70A	100/1A	100/1A	70A	70A	250A	70A	70A	70A
EOL	EOL	EOL	EOL			EOL	EOL		EOL	EOL	
					FUTURE VSD						
		JB	JB					630A FEEDER TO F72-SW-172 2000-F72-SLD-EL-10211	JB	JB	
M 3~	M 3~	M 3~	M 3~	M 3~	M 3~	M 3~	M 3~		M 3~	M 3~	M 3~
0.75kW	0.75kW	3.7kW	3.7kW	75kW	75kW	5.5kW	5.5kW		2.2kW	2.2kW	2.5kW
F72-AG-658 SLAKED LIME DISCHARGE HOPPER 1 AGITATOR 2000-F72-SCM-EL-10668	F72-AG-659 SLAKED LIME DISCHARGE HOPPER 2 AGITATOR 2000-F72-SCM-EL-10669	F72-AT-411 LIME SILO 1 ACTIVATOR 2000-F72-SCM-EL-10651	F72-AT-412 LIME SILO 2 ACTIVATOR 2000-F72-SCM-EL-10652	F72-BW-412 LIME DELIVERY BLOWER 2000-F72-SCM-EL-10665	F72-BW-413 LIME DELIVERY BLOWER (FUTURE) 2000-F72-SCM-EL-xxxxx	F72-FS-411 SCREW FEEDER 2000-F72-SCM-EL-10654	F72-FS-412 SCREW FEEDER 2000-F72-SCM-EL-10656		F72-FV-411A LIME SILO 1 VIBRATING CONVEYOR DRIVE 1 2000-F72-SCM-EL-10655	F72-FV-411B LIME SILO 1 VIBRATING CONVEYOR DRIVE 2	F72-FV-412A LIME SILO 2 VIBRATING CONVEYOR DRIVE 1 2000-F72-SCM-EL-10663	F72-FV-412B LIME SILO 2 VIBRATING CONVEYOR DRIVE 2	F72-CR-660 LIME SLAKING MILL BALL KIBBLE HOIST 2000-F72-SCM-EL-10664

NOTES

1. EARTH-NEUTRAL LINK SHALL BE INSTALLED IN INCOMER TIER. SIGNAGE SHALL BE PLACED ON MCC AND ON TRANSFORMER READING "M.E.N. LINK LOCATED IN F72-SW-072 INCOMER TIER" CABLE SHALL BE IDENTIFIED C/W LABEL STATING "M.E.N. LINK - DO NOT REMOVE".
2. SWITCHBOARD SHALL BE EXTENSIBLE AND BE PROVIDED WITH THE FACILITY TO EXTEND THE MAIN BUSBAR AND APPEND ADDITIONAL TIERS AT ONE END WITHOUT MAJOR STRUCTURAL DISASSEMBLY.
3. UPSTREAM TRANSFORMER FEEDER SHALL BE TRIPPED ON ACTIVATION OF SWITCHROOM EMERGENCY PUSHBUTTON, DETECTION OF EARTH FAULT OR ARC FAULT DETECTION.
4. SWITCHBOARD VENDOR SHALL SELECT SWITCHGEAR TO PROVIDE TYPE-2 CO-ORDINATION SUITABLE FOR HIGH EFFICIENCY MOTORS.
5. ALL DRIVE I/O VIA COMMS NETWORK TO DCS. FOR COMMUNICATIONS BLOCK DIAGRAM REFER 2000-F72-BLK-IC-10001.
6. SURGE DIVERTERS SHALL PROVIDE FEEDBACK TO DCS VIA VOLT-FREE CONTACTS. SURGE DIVERTER FUSES SHALL BE SIZED BY VENDOR TO SUIT DUTY.
7. MINIATURE CIRCUIT BREAKERS SHALL BE SIZED AND RATED TO SUIT ACB AND INCOMER REQUIREMENTS.
8. THERMAL OVERLOAD UNITS MAY BE SUBSTITUTED WITH ELECTRONIC (MAGNETIC) OVERLOAD UNITS TO SUIT DUTY.
9. TOLs TO BE MOUNTED IN PURPOSE-BUILT ENCLOSURE BENEATH VSD. TOLs MAY BE INCLUDED IN MCC MODULE WITH ENGINEERING APPROVAL.
10. ANY DRIVE RTDs SHALL BE WIRED TO DEDICATED REMOTE I/O CARDS IN THE FIELD. RTD RELAYS MAY BE INCLUDED IN THE MCC MODULE WITH ENGINEERING APPROVAL.
11. CIRCUIT BREAKER TO BE SUBSTITUED FOR FUSED-SWITCH WHERE REQUIRED BY EQUIPMENT MANUFACTURERS GUIDELINES.
12. EMERGENCY SHUTDOWN (E-SHUT) PUSH BUTTON WIRED TO SHUNT TRIP NOMINATED CIRCUIT BREAKERS.

LEGEND

- DIRECT ON-LINE MOTOR STARTER GENERIC - NOTE 4, 5
- DIRECT ON-LINE MOTOR STARTER REVERSING - NOTE 4, 5
- EOL — ELECTRONIC OVERLOAD UNIT NOTE 4, 5
- THERMAL OVERLOAD UNIT NOTE 8
- VARIABLE SPEED DRIVE NOTE 4, 5
- EARTH LEAKAGE RELAY NOTE 4, 5
- FIELD-MOUNTED DECONTACTOR
- JB — JUNCTION BOX
- MOTOR THERMISTOR (ONLY REQ. FOR MOTORS >37kW - 250kW)
- RTD — MOTOR RTD (ONLY REQ. FOR MOTORS >250kW)
- MOTOR HEATERS (ONLY REQ. FOR MOTORS >90kW)
- 49 63 71 — TRANSFORMER PROTECTIVE DEVICES (REFER ANSI STANDARDS FOR DESCRIPTIONS)
- CIRCUIT BREAKER (NOTE 11)
- xxxA / xxxA — ST TRIP SET POINT / CB SIZE
- AFD — ARC FLASH DETECTION
- ECR — EARTH CONTINUITY RELAY
- E-SHUT — SWITCHROOM EMERGENCY SHUTDOWN
- 63A 3 PHASE SOCKET OUTLET

2000-F72-SLD-EL-10201_Rev2

F72-SW-072 415V MCC - single line diagram sheet 1

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